

Ex4-2

才子桓 2212700 密码科学与技术

实验要求

调研一种证书

要求：

1. 给出证书本身的文件
2. 指出证书每个字段代表什么

实验过程

这里我选用的证书为 `baidu.com` 的X.509数字证书。

[证书文件](#)

证书获取



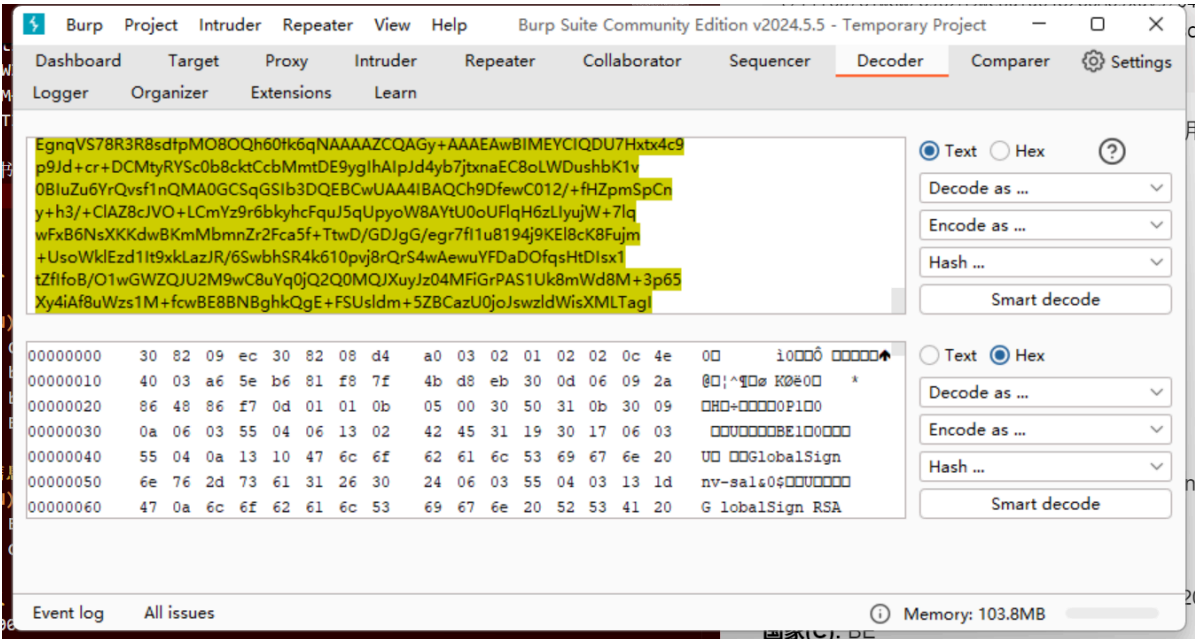


文件内容

```
-----BEGIN CERTIFICATE-----
MIIJ7DCCNSgAwIBAgIMTkADp162gfh/S9jrMA0GCSqGSIb3DQEBCwUAMFAXCzAJ
BgNVBAYTAKJFMRkwFwYDVQQKEgBHBG9iYXNpdWduIG52LXNhMSYwJAYDVQQDEh1H
bG9iYXNpdWduIFJ1TSBPVjBTU0wgQ0EgMjAxODAwFw0yNDA3MDgwMTQxMDJhFw0y
NTA4MDkwMTQxMDZAMFAMQSwCQYDVQQGEwJDTjEQMA4GA1UECBMHYmVpam1uZzEQ
MA4GA1UEBxMHYmVpam1uZzE5MDcGA1UEChMwQmVpam1uZyBCYXNpdWduSjB2b2g
U2NpZW5jZSB1ZWNobm9sb2d5IENVLiwgTHRkMRIwEAYDVQQDEW1iYXNpdWduSjB2b2g
ggEiMA0GCSqGSIb3DQEBAQUAA4IBDwAwggEKAoIBAQC1wFMskJ2dseOqoHptNwot
FOhdBERSz4VQnRNKXEEXMQEfgbNtScQ+C/Z+IprAt1EobhYl1fn74kt2nTscQLng
jFQkrVBuO/6PNGkd1CYGBeGqAL7xR+LoyHnph9mwcBjc+wvt2ZYM9I1c1pXCJa+I
tsq6qpb1AGoQxRDZ2n4K8Gd61wgNCPHDHC/Lk9NPJoUBMvYwVEe51KhHsJtWtHe4
QC3y58Vi+r5R0Pwn2hyTBr9fCo58p/stDiRqp9Irtmi95YhwkNkmgwpmB8RhcGoN
h+Uw5TkPZVj4AVaoPT1ED/GMKZev0+ypmp0+nmjVg2x7yUfLUfp3x7oBdI4TS2hv
AgMBAAGjgggATMIIGjzAOBgNVHQ8BAf8EBAMCBaAwDAYDVR0TAQH/BAIwADCBjgYI
```

KwYBBQUHAQEegYEWfzBEBggrBgEFBQCwAoY4aHR0cDovL3N1Y3VyZS5nbG9iYWxz
awduLmNvbS9jYWN1cnQvZ3Nyc2FvbnNzbGNhMjAxOC5jcnQwNwYIKwYBBQUHMAGG
K2h0dHA6Ly9vY3NwLmdsb2JhbHNpZ24uY29tL2dzcnNhb3Zzc2xjYTIwMTgwVGYD
VR0gBE8wTTBBBgkrBgEEAAyARQWNDAYBggrBgEFBQCcARYmaHR0cHM6Ly93d3cu
Z2xvYmFsc2lnbi5jb20vcMvVb3NpdG9yeS8wCAYGZ4EMAQICMD8GA1UdHwQ4MDYw
NKAyoDCGLmh0dHA6Ly9jcmwuZ2xvYmFsc2lnbi5jb20vZ3Nyc2FvbnNzbGNhMjAx
OC5jcmwwgNhbG9vHREAggNYMIIDVlIjYmFpZHUuY29tggxiYw1mdwJhby5jb22C
DHD3dy5iYw1kds5jboIQd3d3LmJhawR1LmNvbS5jboIPbWN0LnkubnvbwkuY29t
ggthcG9sbG8uYXV0b4IGZHD6LmNuggsqLmJhawR1LmNvbYIOKi5iYw1mdwJhby5j
b22CESouYmFpZHVzdGZ0awMuY29tgg4qLmJkc3RhdG1jLmNvbYILKi5iZG1tzy5j
b22CDCouAGFvMTizLmNvbYILKi5udw9tas5jb22CDSouY2h1Yw5rZS5jb22CDSou
dHJ1c3Rnby5jb22CDyouYmN1LmJhawR1LmNvbYIqKi5lexVuLmJhawR1LmNvbYIP
Ki5tYXAuYmFpZHUuY29tgg8qLm1iZC5iYw1kds5jb22CESouZmFuewkuYmFpZHUu
Y29tgg4qLmJhawR1YmN1LmNvbYIMKi5taxBjZG4uY29tghAqLm5ld3MuYmFpZHUu
Y29tgg4qLmJhawR1cGNzLmNvbYIMKi5haxBhZ2UuY29tggsgLmFpcGFnZS5jboIN
Ki5iY2Vob3N0LmNvbYIqKi5zyWZ1LmJhawR1LmNvbYIOKi5pbs5iYw1kds5jb22C
EiouYmFpZHVjb250ZW50LmNvbYILKi5kbG51bc5jb22CCyouZGxuZWwub3JnghIq
LmR1ZXJvcy5iYw1kds5jb22CDiouc3UuYmFpZHUuY29tgggqLjKxLmNvbYISKi5o
YW8xMjMuYmFpZHUuY29tgg0qLmFwb2xsby5hdXRvghIqLn1ZXNods5iYw1kds5j
b22CESouYmouYmFpZHVjY2UuY29tghEqLmd6LmJhawR1YmN1LmNvbYIOKi5zbWfy
dGFwCHMuY26CDSouYmR0anJjdi5jb22CDCouAGFvMjIyLmNvbYIMKi5oYw9rYw4u
Y29tgg8qLNBhZS5iYw1kds5jb22CESoudmQuYmRzdGF0awMuY29tghEqLmNsb3Vk
LmJhawR1LmNvbYISY2xpY2suaG0uYmFpZHUuY29tghBsb2cuaG0uYmFpZHUuY29t
ghBjbs5wb3MuYmFpZHUuY29tghB3bi5wb3MuYmFpZHUuY29tghR1cGRhdGUucGFu
LmJhawR1LmNvbTAdBgNVHSUEfjAUBggrBgEFBQCDAQYIKwYBBQUHAWIWHwYDVR0j
BBgwFoAU+O9/8s14Z6jeb48kjYjxhwMCS+swHQYDVR00BBYEFK3KAFTK2OWuto+D
2ieAKE5ZJDsYMIIBfwYKKwYBBAHweQIEAgSCAW8EggFrAwkAdgCVGBoo1oyj4Kmk
TJxnqwn4u7wiuq68stijoZ3T+byDDQAAAZCQAGZAAAEAWBHMEUCIFwF5Jc+zyIF
GnpXchz9fY1qz1qg/ovrs2nnuxcpBuuIAiEAu3scD6u51VOP/9aMSqr2yKHZLbHw
Fos9U7AzsdLIZA8AdgAS8U40vVNYTIQGGCOPP3oT+oe1YoeInG0wBYTr5YYm0gAA
AZCQAG3iAAAEAWBHMEUCIBBYQ6NP7VUDgfktWRg5QxT23QAbTqYovtV2D9O8Qc0T
AiEA2P7+44EvQ5adwL1y56oyxv/+Gujeia7wpo7+Xbhv6MAdwAN4fiwk9MNwUBi
EgnqVS78R3R8sdfpM08Oqh60fk6qNAAAAZCQAGY+AAAEAWBIMEYCIQDU7Hxtx4c9
p9Jd+cr+DCMtyRYSc0b8cktCcbMmtDE9ygIhAipJd4yb7jtxnaEC8oLWDushbK1v
0BIuZu6YrQvsf1nQMA0GCSqGSIB3DQEBCwUAA4IBAQch9DfewC012/+fHZpmSpCn
y+h3/+c1AZ8cJVO+LCmYz9r6bkyhcFquJ5qUpYow8AYtU0ouF1qh6zLIyujw+7lq
wFXB6NsXKKdwBKMbmZr2Fca5f+TtwD/GDJgG/egr7fi1u8194j9KE18ck8Fujm
+Usowk1Ezd1It9xkLazJR/6SwbhSR4k610pvj8rQrS4wAewuYFDaDOfsHtDIsx1
tzfIfOB/O1wGWZQJU2M9wc8uYq0jQ2Q0MQJXuyJz04MFiGrPAS1uk8mwd8M+3p65
xy4iAf8uWzs1M+fcwBE8BNBghkQgE+FSus1dm+5ZBCazU0joJswz1dwisXMLTagI
-----END CERTIFICATE-----

X.509 数字证书中间使用 Base64 编码加密，用 Burpsuite 对加密部分解密：



字段分析

主题信息

版本: V3
通用名称(CN): baidu.com
国家(C): CN
省份(S): beijing
城市(L): beijing
组织(O): Beijing Baidu Netcom Science Technology Co., Ltd

签发者信息

通用名称(CN): GlobalSign RSA OV SSL CA 2018
国家(C): BE
组织(O): GlobalSign nv-sa

证书信息

序列号: 000000004e4003a65eb681f87f4bd8eb
类别归档: OV
证书类型: 服务器证书
用途: serverAuth,clientAuth
密钥类型: RSA
密钥强度: 2048
签名算法: SHA256
颁发时间: 2024-07-08 09:41:02
过期时间: 2025-08-09 09:41:01
SHA-256指纹证书:
9073ded9d993a934c29c5ec3c6afa7286d2f0f8848352f94d02035865d8568e2
SHA-256指纹公钥:
83c0ac75ca7200ac662e8585bcc776842ed90d4cbb2f81363583a2d62f2a12d1
签名信息:

a1f437dec02d35dbff9f1d9a664a90a7cbe877ffe0a5019f1c2553be2c2998cf
dafa6e4ca1705aae279a94a72a16f0062d534a14165a87eb32c8cae8d6fbb96a
c05c41e8db1728a77004a98c6e69d9af615c6b97fe4edc03fc60c9806fde82be
df235bbcd7de23f4a125f1c2bc16e8e6f94b285a4944cddd48b7dc642dacc947
fe92c1b85247893ad74a6f8fcad0ad2e3001ec2e6050da0ce7eab07b4322cc75
b597c87e807f3b5c0659940953633dc02f2e62ad23436434310257bb2273d383
05886ac012d5493c99677c33ede9eb95f2e2201ff2e5b3b3533e7dcc0113c04
d06086442013e15252c95d9bee590426b35348e826cc3395d5a2b1730b4da808

备用名:

baidu.com
baifubao.com
www.baidu.cn
www.baidu.com.cn
mct.y.nuomi.com
apollo.auto
dwz.cn
*.baidu.com
*.baifubao.com
*.baidustatic.com
*.bdstatic.com
*.bdimg.com
*.hao123.com
*.nuomi.com
*.chuanke.com
*.trustgo.com
*.bce.baidu.com
*.eyun.baidu.com
*.map.baidu.com
*.mbd.baidu.com
*.fanyi.baidu.com
*.baidubce.com
*.mipcdn.com
*.news.baidu.com
*.baidupcs.com
*.aipage.com
*.aipage.cn
*.bcehost.com
*.safe.baidu.com
*.im.baidu.com
*.baiducontent.com
*.dlne1.com
*.dlne1.org
*.dueros.baidu.com
*.su.baidu.com
*.91.com
*.hao123.baidu.com
*.apollo.auto
*.xueshu.baidu.com
*.bj.baidubce.com
*.gz.baidubce.com
*.smartapps.cn
*.bdtjrcv.com
*.hao222.com

```
*.haokan.com
*.pae.baidu.com
*.vd.bdstatic.com
*.cloud.baidu.com
click.hm.baidu.com
log.hm.baidu.com
cm.pos.baidu.com
wn.pos.baidu.com
update.pan.baidu.com
```

ca_url: <http://secure.globalsign.com/cacert/gsrsoavsslca2018.crt>

ocsp_url: <http://ocsp.globalsign.com/gsrsoavsslca2018>

crl_url: <http://crl.globalsign.com/gsrsoavsslca2018.crl>

公钥

公钥:

```
-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAtcBTLJCdnBHjqqB6bTcK
LRTToXQREbGeFUJ0TS1xBFzEBH4GzbUnEPgv2fiKUQLdRdM4WJYn5++JLdp07AkC5
4I30JEVQbjv+jzRinZQmBgXhggC+8Ufizsh56R/ZsAgSXP1lbdS2DPSNXJavwiwv
iLbKuqqw9QBqEMUQ2dp+CVBnetCIDQjxwx3Py5PTTyafATL2FrXHuzSoR7CbvrR3
uEAt8ufFYvq+UdD1p9ockwa/XwqOfKf7LQ4kaqfSK7ZoveWicJDZJoMKTAfEYXBq
DYf1MOU5D2VY+AFWqD09RA/xjCmXr9PsqZqdPp5o1YNse81Hy1H6d1+6AXSOE0to
bwIDAQAB
-----END PUBLIC KEY-----
```

PN(e): 65537

PN(n):

```
2294397214112921827346324144711638177008927085453744230808308136
1320209018936360080745206594460397244069334081731261674797287927
9503606498439532864961131042342900066918409887945907568523478391
6251985880037389079691515673857167120747755145039839059681125971
9157842620865920572333853700320587081895115675295752201668184969
8488803874901944434278354577617591114909173067867655989729472374
7262736339673632839973806939801061061528345746184164668416440642
5771427656087598671248603006266849585794944596926887801649432098
9500765003697460867873694274064058100096082650387734030705770096
21896810087449625205031486346569471191151
```